

Programme	B.Sc Computer Science				
Course Title	Theory of Computation				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Understanding of basic mathematical concepts such as sets, functions, relations, logic and discrete structures. 2. Understanding of fundamental programming constructs such as loops, conditionals, functions, and recursion.				
Course Summary	This course covers a comprehensive exploration of fundamental concepts in computer science, delving into computational models, formal language theory, and computational complexity. Students learn about various computational models such as finite automata, pushdown automata, and Turing machines, gaining insights into their capabilities and limitations. Through the study of formal languages and grammars, students understand the structure and properties of regular and context-free languages.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To learn and understand fundamental concepts in computational theory, including computational models such as finite automata, pushdown automata, and Turing machines.	U	P	Practical Assignment / Instructor-created exams / Quiz
CO2	To be able to classify formal language into regular, context-free, context sensitive and unrestricted languages.	Ap	P	Practical Assignment / Instructor-created exams / Quiz
CO3	To design and analyse Turing machines, their capabilities and limitations	Ap	P	Practical Assignment / Instructor-created exams / Quiz
CO4	Construct the abstract machines including finite automata, pushdown automata, and Turing machines from their associated languages and grammar	Ap	P	Practical Assignment / Instructor-created exams / Quiz
CO5	Gain insights into decidability and undecidability, and understand the limitations of computation through the study of the halting	Ap	P	Practical Assignment / Instructor-

	problem and other undecidable problems.			created exams / Quiz
CO6	Solve computational problems regarding their computability and complexity and prove the basic results of the theory of computation	E	P	Practical Assignment / Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Theory of Computation

Module	Unit	Content	Hrs	Max Marks
I	FINITE AUTOMATA		10	16
	1	Formal Language: Definition, Chomsky classification of Grammar, Language and Relation, Language and Automata	2	
	2	Finite Automata: DFA, NFA with and without ϵ - moves	2	
	3	Equivalence of DFA and NFA	2	
	4	Equivalence of NFA and ϵ -NFA	2	
	5	Mealy and Moore Models - Conversions	2	
II	REGULAR LANGUAGE, REGULAR EXPRESSION		10	18
	6	Regular Languages: Regular Expressions, Ardens Theorem	2	
	7	Conversion of Regular Expression to Finite Automata	2	
	8	Closure properties of RLs	2	
	9	Pumping lemma for RLs	2	
	10	Myhill-Nerode theorem	2	
III	PUSH DOWN AUTOMATA, CONTEXT FREE LANGUAGE		14	18
	11	Pushdown Automata - Instantaneous Description - Transition Diagram	3	
	12	Deterministic and Non Deterministic PDA	3	
	13	Equivalence of PDAs and CFGs, Pumping lemma for CFLs	2	
	14	Closure properties of CFLs, Simplification of CFLs	2	

	15	Chomsky Normal form (CNF) and Greibach Normal form (GNF)	2	
	16	CYK algorithm for CFL membership	2	
IV	TURING MACHINE, UNDECIDABILITY		14	18
	17	Closure properties of CFLs, Simplification of CFLs	3	
	18	Chomsky Normal form (CNF) and Greibach Normal form (GNF)	3	
	19	CYK algorithm for CFL membership	2	
	20	Church Turing hypothesis - Rices theorem	2	
	21	Undecidability of Posts correspondence problem	2	
	22	The Class P and NP	2	
V	Open Ended Module- Application Level		12	
		1. Application of regular expressions in pattern matching and text processing. 2. Analysis of context-free languages using pumping lemma and closure properties. 3. Investigation of undecidability and un-solvability using the halting problem and Rice's theorem. 4. Notion of tractability: The Class P and NP , NP completeness of propositional satisfiability, other variants of satisfiability. NP-complete problems from other domains:		
		graphs (clique, vertex cover, independent sets, Hamiltonian cycle), number problem (partition), set cover. 5. Discussion of practical implications and applications of complexity theory.		

Textbooks

1. J.E.Hopcroft, R.Motwani and J.D Ullman, —Introduction to Automata Theory, Languages and Computations, Third Edition, Pearson Education, 2016.
2. Theory of Computer Science- Automata, Languages and Computation- K.L.P. Mishra, N Chandrasekaran, PHI

Reference books :

1. Theory of Computation, Sachin Agrawal, Vikas Publishing House

2. Micheal Sipser, —Introduction of the Theory and Computation, Thomson Brokecole, 3rd Edition, 2013.
3. J.Martin, —Introduction to Languages and the Theory of Computation, Third Edition, TMH, 2007. .
4. An Introduction to the Theory of Computer Science, Languages and Machines-Thomas A. Sudkamp, Third Edition, Pearson Education.
5. An Introduction to Formal languages and Automata- Peter Linz

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	3	2	1	-	-						
CO 2	2	1	2	1	-	-						
CO 3	1	3	1	1	-	-						
CO 4	-	3	3	2	-	-						
CO 5	-	1	3	3	1	-						
CO 6	-	1	3	3	1	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓